Karthik Reddy Basupally

+91 9063007185 - basupallykarthikreddy@gmail.com - linkedin - github - leetcode

EDUCATION

VIT University **Vellore, Tamil Nadu, India**
B.Tech in Electronics & Communication Engineering, CGPA-8.71/10.0 *Sep 2020 — May 2024*

TECHNICAL SKILLS

Languages: Java (8–21), SQL, Python, TypeScript, JavaScript, Bash
Backend: Spring Boot, JPA, Apache Camel, RESTful APIs, OpenAPI/Swagger, Microservices Architecture
Databases: MS SQL Server (views, stored procedures, query tuning), PostgreSQL, SQLite
DevOps & Tools: Docker, GitLab CI/CD, Jenkins, Maven, Git, Grafana, SonarQube, Jira
Frontend: React.js, TypeScript, Tailwind CSS
Concepts: Distributed Systems, Event-Driven Processing, Concurrency & Multithreading, Batch Processing, Transaction Management, Caching Strategies, Performance Optimization, Object-Oriented Programming (OOP).

WORK EXPERIENCE

Software Engineer
Nomura, Mumbai, India *Jul 2024 — Present*

- Built high-throughput, low-latency backend systems in Global Markets IT (Prime Brokerage) for regulatory reporting, client reporting, trade reconciliation, and enterprise integration.
- Developed a **Spring Boot** regulatory trade persistence microservice processing **60M+ trade records**; optimized batching, transaction management, and JPA persistence to reduce insertion latency from 3s to 40ms.
- Served as **SME** for **4+** business-critical **Java** and **Apache Camel** adapters across workflows; owned production support, incident triage, debugging, releases, and deployments.
- Built a reusable integration gateway framework using the **Template Design Pattern** with configurable routing, transformations, centralized exception handling, Grafana metrics, and health checks; benchmarked at 462 HTTP messages/sec and adopted across multiple services.
- Integrated Java and Apache Camel adapters with an external vendor's AWS-based data feeds — polling SQS queues over HTTP and downloading files directly from S3, with validation checks before downstream processing.
- Built a **Python** automation framework integrating the **Jira** and **GitLab** APIs to automate dependency upgrades, multi-repository code changes, commits, and Merge Requests across **20+ Spring Boot microservices**.

Software Engineer Intern
Wizergos, Bengaluru, India *May 2023 — Jul 2023, Dec 2023 — Mar 2024*

- Re-engineered a customer-facing chatbot from **AngularJS** to **React.js**, leveraging Hooks, component-based architecture, and state management to improve maintainability and user experience.
- Improved chatbot response performance by **50%** through frontend optimization and efficient API integration.

PROJECTS

- **SonarAutoFix – LLM-Powered CI/CD Remediation (Nomura)** - Conceptualized and built an AI remediation platform using Dockerized GitLab Runners, LLM workflows, and Javalang parsers to analyze SonarQube findings, generate code fixes, and raise Merge Requests for developer review; remediated 40% of SonarQube issues across enterprise microservices, recognized with the **Nomura Star / Quarterly Recognition Award**.
- **CodeRAG – Code-Aware Retrieval-Augmented Generation (RAG)** - Developed a code-aware RAG system that models source code using method signatures, call graphs, static analysis, and vector embeddings instead of flat-text chunking, enabling semantic code search and conversational querying through a FastAPI backend using Python, Sentence Transformers, ChromaDB, React, and TypeScript. GitHub

EXTRACURRICULAR ACTIVITIES

- Earned 5+ skill badges/certifications from industry-recognized platforms (IBM, Cisco, Google, AWS).
- Led and organized weekly coding sessions for the Tech Enthusiasts Club, fostering skill development and a collaborative learning environment.